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Key Points:

- Lidar differencing reveals range-wide volumes of scour and re-deposition by post-wildfire debris flows within Santa Ynez Range canyons
- Scoured coarse-grained sediment, emplaced by prior colluvial and debris-flow processes, constituted ~70% of the entire debris-flow volumes
- Volumes of sediment available for debris-flow scour depend on the stochastic interaction of accumulation processes and large scouring events

Supporting Information:

Supporting Information may be found in the online version of this article.

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Sediment Recruitment and Redistribution in Mountain Channel Networks by Post-Wildfire Debris Flows

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Abstract We used lidar differencing and field observations to map volumes, and interpret the origins of, sediment mobilized from mountain canyons by large post-wildfire debris flows near Montecito, CA, USA in 2018. The debris flows progressively entrained and partially redeposited 550,000 m^3 of previously stored channel sediments throughout the canyon networks. The observations that scour depths and volumes were highest where the largest volumes of bouldery colluvium and debris-flow deposits had accumulated, and that scour persisted beyond the mountain front, indicates that debris-flow volumes in this extreme event were ultimately controlled by the coarse sediment reservoir available for scour. Because the volumes of available stored sediment result from the stochastic interaction of colluvial mass wasting, the magnitude and frequency of previous debris flows, and the accommodation space provided by valley morphology, the study reinforces the importance of estimating stored sediment volumes when developing debris-flow hazard assessments.

Plain Language Summary We measured the volumes, and identified the origins of, sediment evacuated from a mountain range by large debris flows near Montecito, CA, USA in 2018. We measured sediment volumes by subtracting the ground surface topography of creek bottoms estimated using lidar scanning equipment before and after the events. We conducted field campaigns to investigate the type of sediment removed during the debris flows. The flows evacuated and partially redeposited a total of 550,000 m^3 of material, with the greatest removal occurring in large canyons where previous boulder-rich debris-flow deposits had accumulated. Our results indicate that estimates of stored sediment are necessary when assessing debris-flow hazard in similar high-value environments.

1. Introduction

Understanding the origin and volumes of sediment supporting debris-flow growth is important for interpreting processes that erode mountains (e.g., Stock & Dietrich, 2006; Tucker & Hancock, 2010), and for constructing runout models for hazard delineation (e.g., Berger et al., 2011; Frank et al., 2015; Guthrie et al., 2010; Rickenmann, 1999; Wieczorek et al., 2001). Debris flows are initiated either by hillslope failures at discrete sites (e.g., Campbell, 1975; Stock & Dietrich, 2006), by erosion of sediment from headwater channels (McCoy et al., 2010; Prancevic et al., 2014), or by accelerated surface erosion on hillslopes (Alessio et al., 2021; Cannon et al., 2003; Wall et al., 2020). As debris flows travel downstream, they grow by incorporating sediment from valley floors, often increasing to many multiples of their original size before deposition on fans and floodplains (Gabet & Bookter, 2008; Reid et al., 2016; Santi et al., 2008). The volume, discharge rate, and granulometry of flows control their damage potential (e.g., Kean et al., 2019). However, the roles of stored sediment and channel morphology in sediment recruitment and deposition during debris-flow events remain poorly understood, especially for large flows in steep landscapes following wildfire.

This lack of understanding occurs in part because measurement of the spatial distribution of material scoured and deposited during large post-wildfire debris flows requires obtaining high-resolution topographic data for infrequent events. Early attempts to measure sediment recruitment by debris flows were constructed by field surveys of channel cross-sections (e.g., Gabet & Bookter, 2008; Jakob et al., 2005; Santi et al., 2008). Recent studies have relied on airborne lidar or photogrammetry to measure topographic change at the centimeter scale (e.g., Ellett et al., 2019; Guilinger et al., 2020; McGuire et al., 2017; Pelletier & Orem, 2014; Rengers et al., 2021). For example, DiBiase and Lamb (2020) used a 5-year time series of airborne lidar surveys to show that sediment accumulated (~3 m) by dry ravel in headwater channels is a primary contributor to rainfall-induced debris flows in burned, bedrock-dominated landscapes. However, there have been few opportunities to measure spatially

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continuous vertical change resulting from post-wildfire debris flows at the scale of entire mountain catchments. Here, we use differential lidar measurements at the submeter-scale to measure volumes and spatial patterns of debris-flow scour and deposition in a 2018 debris-flow event near Montecito, CA, USA, that spanned an entire mountain range (>25 km^2), from ridgeline to fans. We couple the lidar differencing with field observations to investigate the nature, magnitude, and distribution of sediment recruitment and deposition during the debris flows, and to interpret how channel sediment accumulates to favor the creation of large, damaging debris-flow events.

2. Montecito Debris Flows of 2018: Study Area and Debris-Flow Initiation

The debris flows that devastated Montecito, CA, USA, on January 9, 2018 were activated by a brief, intense rainstorm in the Santa Ynez mountains (Kean et al., 2019; Lancaster et al., 2021; Oakley et al., 2018). Only three weeks earlier, the 2017 Thomas Fire resulted in the near-complete removal of vegetation on the steep mountain hillslopes at moderate-high soil burn severity, which generated partially hydrophobic soil conditions (USDA Forest Service, 2018). During the rainstorm, slurry generated by extensive rill erosion of hillslope colluvium (Alessio et al., 2021) entered channel networks, leading to the mobilization of dry ravel and boulder-rich sediment, previously supplied by colluvial and debris-flow processes (Kean et al., 2019; Lancaster et al., 2021). The debris flows scoured nearly the entire channel networks of six adjacent mountainous catchments from the ridgetops (elevation ~1,000 m) to the fans (elevation ~150 m; Figure 1a and Table S1 in Supporting Information S1). These watersheds drain a series of alternating sandstone and shale formations (Figure 1a): Juncal Shale at the ridgeline, the Matilija Sandstone at mid-elevations, and the Cozy Dell Shale and Coldwater Sandstone at the lowest elevations. The Juncal Shale Fm contains m-thick lenses of sandstone bedrock that become more massive and extensive to the east.

3. Methods

3.1. Lidar Differencing

To measure topographic change due to the debris flows, we created a DEM of Difference (DoD) by subtracting elevations between bare-earth DEMs collected in 2015 (pre-debris flow; Morell et al., 2021) and 2018 (post-debris flow; CDGS, 2018). The DoD was produced following creation of bare-earth DEMs from point clouds, co-registration of the two DEMs, and clipping to a channel support area of 0.001 km^2. The final DoD was produced after excluding outliers, artifacts, and DoD data below a minimum level of detection ($_{min}LoD$) of 0.2 m (Schwanghart & Scherler, 2014; Stoker et al., 2016; Wheaton et al., 2010; Supporting Information). We assessed potential data loss in the DoD by interpreting values across local spatial gaps (≥3m) in the 2015 point cloud. Details of the co-registration, identification of outliers, and estimation of data loss procedures are described in Supporting Information S1.

3.2. Field Observations

During a six-month period beginning three weeks after the debris flows, we conducted surveys of primary channels and some tributaries from ridgeline to mountain front in each of the six largest catchments. We examined the sedimentology and stratigraphy of channel margins to identify the origin of sediments stored in the channels prior to the 2018 event. We also examined deposits from the 2018 event and subsequent mass wasting. The field observations were supplemented from high-resolution (5–30 cm) orthomosaics (Alessio et al., 2021) and the DoD. We measured scour depths in the field to ground-truth the lidar differencing results, using observations such as: undercutting into channel banks and beneath large boulders, large tree roots or stumps ripped apart and protruding from vertical scarps along the bank, and elevations of blue, weathered rock in channel banks, interpreted to indicate the elevation of the antecedent dry-season water table at or below the channel bed.

4. Results

Our results provide a comprehensive, range-wide picture of intra-montane scour and re-deposition during the 2018 debris flows and highlight the important roles of sediment storage and valley morphology in debris-flow growth.

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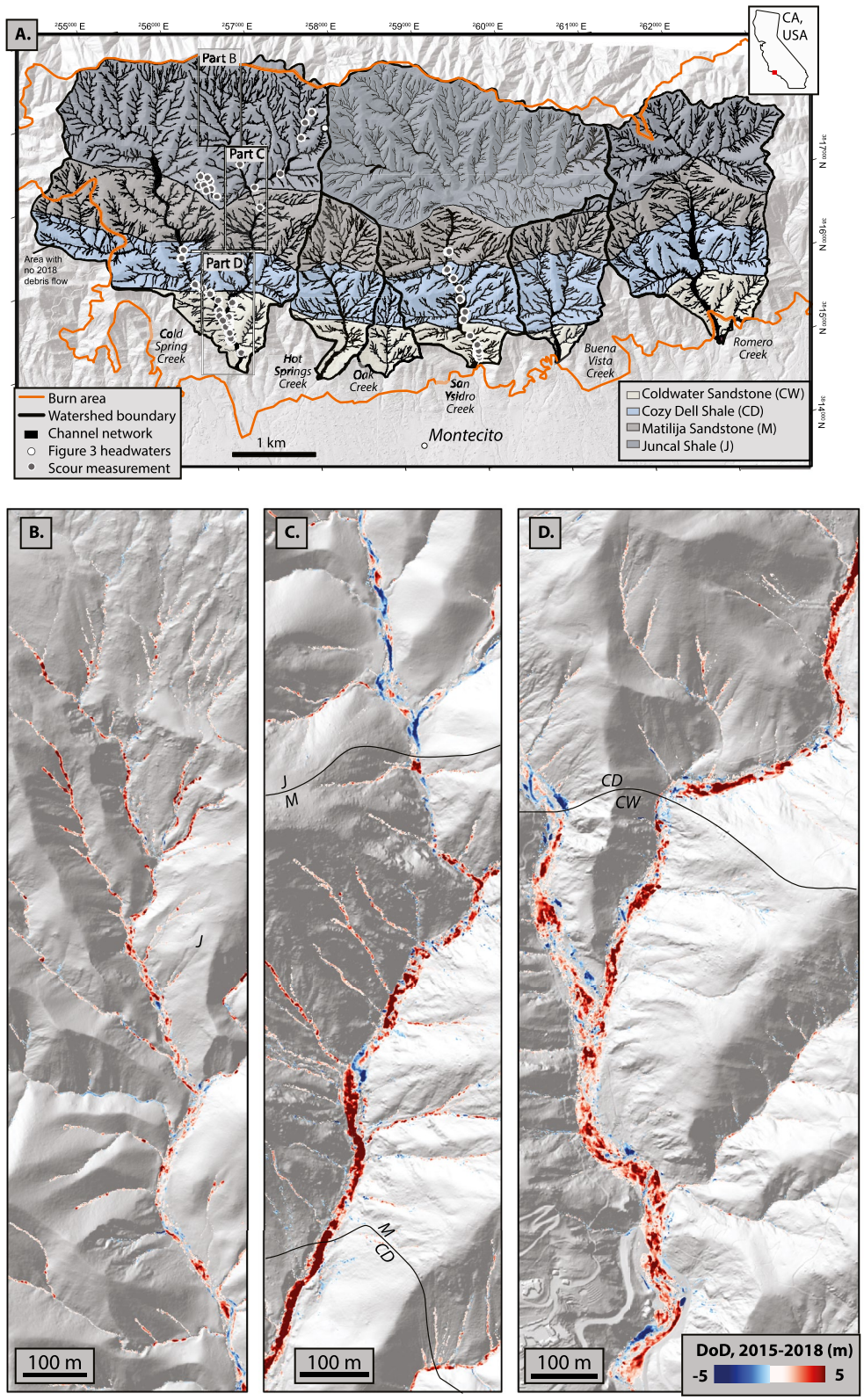


Figure 1. (a) Southern flank of the Santa Ynez Mountains affected by the 9 January 2018, Montecito debris flows. Black outlines indicate extent of lidar differencing. Geologic contacts refined from Dibblee, and Ehrenspeck (1986b) by field observations. Thomas Fire perimeter retrieved from https://rmgsc.cr.usgs.gov/outgoing/GeoMAC/2017_fire_data/California/Thomas/. (b-d) DEMs of Difference (DoD) along the East Fork of Cold Spring Creek. Positive (red) erosion; negative (blue) deposition. See part (a) for locations.

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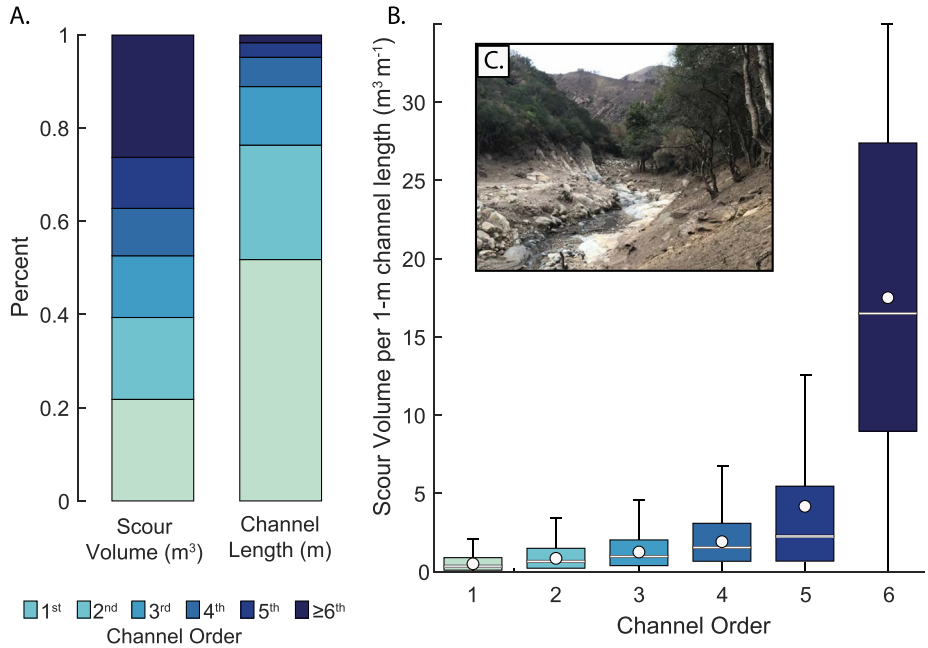


Figure 2. (a) Percent contribution of total scour volume and channel length by channel order for all catchments. (b) Box plots showing mean (white dot), median (bar), and interquartile range (box) of scour volume per meter of channel length. Whiskers show minimum and maximum, excluding outliers (>1.5× interquartile range). (c) Example of scour to bedrock and overbank deposition in San Ysidro Creek.

4.1. Lidar Differencing

4.1.1. Total Volumetric Contributions

The lidar differencing results indicate that the 2018 debris flows scoured and re-distributed a minimum of 550,000 m^3 of sediment from channels within the six mountain catchments with a total drainage area of 25.5 km^2, ~90% of which was within the Thomas Fire perimeter (Figures 1a and S1 in Supporting Information S1). Of this scoured volume, 85% was exported to the piedmont fans (470,000 m^3) and 15% was re-distributed and deposited within the mountain canyons (76,000 m^3) (Figure 2 and Table S1 in Supporting Information S1). Some of the locations where net deposition was recorded could have been scoured before the deposition, so the scour volumes recorded were minimum estimates. First-order channels constitute ~50% of the network length and supplied ~20% of the total scoured volume (100,000 m^3) (Figure 2 and Table S2 in Supporting Information S1). The ≥6th order channels contributed 26% of the scour volume (123,000 m^3), even though they comprise less than 2% of the channel network length (Figure 2).

4.1.2. Downstream Changes in Debris-Flow Scour

Scour depths and volumes per unit of channel length generally increased with distance downstream, and their patterns were affected by channel order, channel morphology, and bedrock lithology. The overall patterns of scour or deposition were largely similar in the three largest watersheds (Cold Spring, San Ysidro, and Romero, Figure 1a) and are represented by the measurements from the DoD along a transect of East Fork Cold Spring Canyon (Figure 1). Along channels from first to fifth order on the Juncal shale, as channel gradient declined from 0.5 to 0.15 and flow width increased from 2 to 20 m, scour depths fluctuated between 0.5 and 2 m. Volumetric recruitment rate fluctuated between ~0.5–1.0 m^3 per meter of channel length, without trend (Figure 3). Scour depth and recruitment rate increased abruptly as the flows entered the Matilija Sandstone outcrop and rose to 4.5 m and 30 $m^3\ m^{-1}$, respectively, at the downstream end of that outcrop. The trends were interrupted only as the channel crossed massive bedrock outcrops (BR in Figure 3). Scour depth continued to fluctuate between 2 and 4 m as the channel traversed the Cozy Dell Shale and Coldwater Sandstone outcrops, where scour rates ranged over 10–35 $m^3\ m^{-1}$. The flow width in these ≥6th-order reaches increased irregularly from ~20 to 70 m. When averaged over all channels and catchments, the recruitment rate rose from 0.5 $m^3\ m^{-1}$ at first order to 17.5 $m^3\ m^{-1}$ at ≥6th order, with a particularly strong increase between fifth and sixth orders (Figure 2b).

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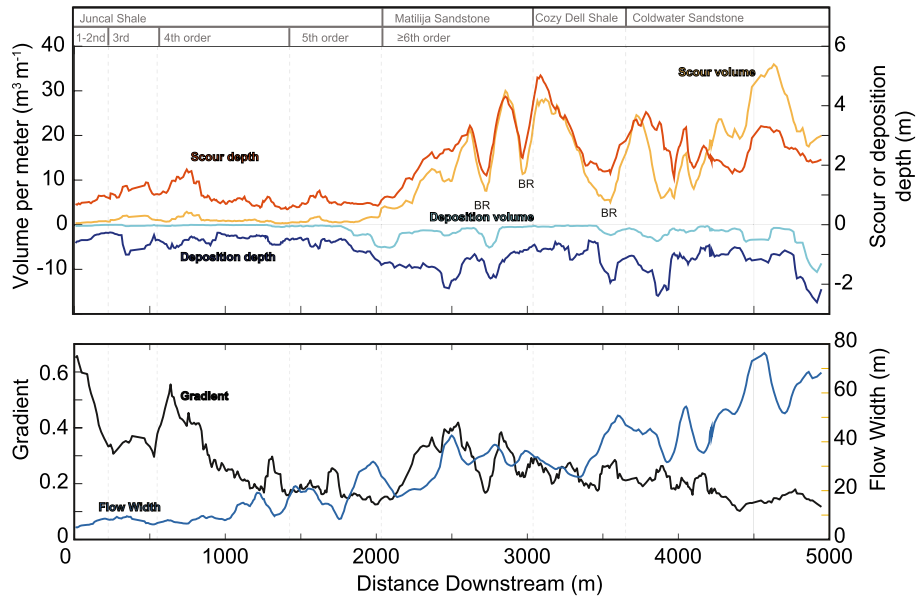


Figure 3. DoD values and channel properties along the East Fork of Cold Spring Creek (Figure 1a). Scour (red) is positive; deposition (blue) is negative. Large channel steps scoured to bedrock (BR). Plots of DoD, flow widths and gradients were smoothed with a 100-m moving average of data sampled every ~10 m.

4.1.3. Deposition

Reach-averaged deposition rates per unit of channel length in East Fork Cold Spring Creek were much lower (0–4 $m^3\ m^{-1}$) than the scour rates, and deposition locally reached maximum depths of ~2 m (Figure 3). Local deposit thickness was generally greater downstream of the Juncal-Matilija contact, as boulders became available for levee building, and as the valley-floor width increased from ~20 to ~70 m (Figure 3). Deposition generally occurred where flow velocities and shear stresses are expected to be relatively low, such as on the inside of bends, where steep tributaries fed sediment into a larger channel, where gradients declined sharply, and where widening valley floors provided accommodation space for marginal deposits (Figures 2c and 4e and 4g).

4.2. Field Observations

Our field observations indicate that the downstream changes in debris-flow scour and growth rate depended strongly on the amount and type of sediment stored in the channel floors prior to the 2018 debris flows.

4.2.1. Sediment Stored in Channel Networks

The first-order channels defined by the GIS-based analysis are consistent with the bedrock channels on sandstones and colluvial channels on shales, as defined by Montgomery and Buffington (1997). Before 2018, these channels on shale units (Figure 1) contained discontinuous 0.3–1 m thick, fine-grained colluvial deposits containing occasional gravel, cobbles, and small sandstone boulders eroded from sandstone lenses within the shale units (Figures 4a and 4b). The sediment was stabilized by woody plants in the steep channels, and fluvial action maintained smaller channels within these colluvial fills. The low-order channels incising sandstone units were either scoured to bedrock before 2018 or contained sparse gravel and sub-angular colluvial boulders (Figure 4a). Channels of third to fifth order were characterized by increasingly large reservoirs of sub-angular to rounded sandstone boulders, averaging 0.75 m in diameter but ranging up to several meters, supplied by Matilija Sandstone outcrops or sandstone lenses within shale units (Figures 4c–4e). The sixth- and seventh-order channels crossing the Cozy Dell Shale and Coldwater Sandstone contained large volumes of subrounded to subangular sandstone boulders, 0.5–6+ m in diameter, but no boulders from shale units (Figures 4f and 4g). The channel surfaces of the boulder-rich reaches had discontinuous covers of sparse subrounded, weakly imbricated boulders (up to ~0.5 m), gravel, and small fractions of sand (Keller et al., 2015), which we interpret to be ephemeral fluvial deposits. Sediment accumulations from post-wildfire dry ravel were observed after the Thomas Fire in some

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Figure 4. Field evidence for scour and the nature of accumulated sediments. (a) First-order channel in the Juncal Shale, showing scour through colluvium to bedrock. (b) Scour through fine-grained alluvium with occasional cobbles and angular boulders from sandstone lenses within the Juncal Shale. (c) Boulder field in fifth-order canyon below the Juncal–Matilija contact. Largest boulders (~4–5 m) were not transported during the 2018 event and are perched on bedrock in the channel. (d) and (e) Scour through debris flow deposits in fifth-order channel. The flows cut narrow channels through the deposits, and deposited boulders above the pre-debris flow channel margin. (f) and (g) Evacuation of clast-supported, antecedent debris-flow deposits in sixth-to seventh-order channels.

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areas (USDA-Forest Service, 2018), but the accumulations that we observed on trails after January 9 were sparse and small. We interpreted most antecedent sediment stores as colluvial rockfall and clast-supported debris-flow deposits. The thickness of these debris-flow deposits (often >3–5 m) suggests that they represent stacked sequences of two or three flow units (e.g., Figure 4d). In sixth and seventh-order reaches, the debris-flow deposits had slightly higher matrix contents.

4.2.2. Scour

The field surveys confirmed that scour was primarily controlled by the amount and nature of the accumulated sediments. We observed that bedrock was exposed over large fractions of most first-order channels. Some channels on shales were locally incised to depths of up to 1–2 m through accumulated colluvium and into friable bedrock, although average scour depths were much smaller. There was little or no evidence of such deep incision in channels on sandstones (Figure 4a). Across most of the landscape, scour in first- and second-order channels was sparse and discontinuous, despite the high gradients. In the wider >4th-order reaches downstream, we observed the 2018 debris flows to have scoured previously accumulated material either to (a) bedrock; (b) a surface consisting of tightly jammed, very large boulders (typically > 4–5m); or (c) a surface no coarser (one to several meters) than the mobilized material but still tightly jammed (Figure 4g). Scour depths and volumes per unit distance were generally lower where massive bedrock outcrops were exposed in the primary channels (see reaches labeled BR in Figure 3).

Although scour into antecedent deposits was pervasive and commonly exposed bedrock, we did not find extensive field evidence of incision into sandstone bedrock, and we observed only minor amounts of chipping or quarrying of sandstone. We did not observe widespread evidence of gouging or grinding of sandstone formations, even where bedrock within the phreatic zone was weathered to a bluish color (Figure 4e). Instead, bedrock channels in the massive sandstone units exhibited flutes and pools, characteristic of fluvial erosion. In the friable shale formations, first to second order channels exhibited U-shaped scoured profiles, indicating some bedrock erosion, but this condition was not observed in higher-order channels.

4.3. Data Quality and Field Validation

Our field observations consistently corroborate the lidar differencing results, and our field measurements of channel scour generally agree with scour estimates from lidar differencing (Figure S2 in Supporting Information S1). The observation that the DoD for the unburned sub-catchment of Cold Spring (Figures 1a and S1 in Supporting Information S1) exhibits much less erosion than adjacent burned watersheds (Figure S1 and Table S1 in Supporting Information S1) suggests that the elevation changes in the DoD accurately represent channel erosion by the 2018 debris flows (Supporting Information S1). Interpolation across data gaps 3 m or larger (Supporting Information S1) suggests that spatial gaps in the 2015 data set account for ∼51,000 m^3 of total erosion and deposition (Table S3 in Supporting Information S1), which represents ∼10% of the total material redistributed during the 2018 event (550,000 m^3; Table S1 in Supporting Information S1). Thus, the volume estimates indicated by the DoD represent minimum approximations of the total volume mobilized within the channel network during the 2018 debris flows.

5. Discussion and Implications

Although sediment recruitment per unit of channel length generally increased downstream in each catchment, as drainage area and slurry discharge from catchment hillslopes increased, the overwhelming influence on the recruitment pattern was the volume of sediment stored in the valley floors prior to 2018. The change in recruitment rate between fifth- and sixth-order channels was not proportional to drainage area, which scaled the volume and presumably discharge rate of slurry from the catchments. The highest rates of sediment recruitment from channels occurred in fifth to seventh-order reaches, where channels were widening and gradients declining, but the debris flows retained the capacity to transport large boulders. Channel-bed scour extended beyond the mountain front onto each fan head. The rate of scour per unit length of channel increased most strongly where we observed abundant accumulation of bouldery colluvium and clast-supported debris-flow deposits within reaches that intersect the Matilija and Coldwater sandstones (Figure 3). Mass wasting on steep (up to 45°) slopes of the two sandstone formations delivered essentially all of the large boulders to the valley floors of the high-order channels, either directly from valley walls, or by debris-flow transport from steep tributaries. Primary channel

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reaches traversing the Cozy Dell Shale were loaded mainly with Matilija Sandstone boulders. The shale formations supplied only minor quantities of small boulders (Figure 4b), although we observed some 1–4 m boulders in paleo-deposits at the mouths of a few tributaries draining the Cozy Dell Shale outcrop. They appear to have originated from sandstone units within the formation close to the Matilija contact. Scour depths in both primary and tributary reaches were limited either by massive bedrock (Figures 2c and 3) or by jammed boulder accumulations (Figures 4e and 4g) too resistant to be moved by even the highly erosive 2018 flows.

The relatively wider and lower-gradient fifth to seventh order reaches provided accommodation space for deposition by earlier, less-powerful debris flows (Figures 2c and 4), which in turn provided the sediment for high recruitment rates during the more powerful 2018 debris flows. These antecedent debris-flow deposits within the relatively low-gradient, high-order channels accumulated in repeated events that, unlike the 2018 flows, left boulder-rich sediment filling the valley floors up to at least 6 m deep. The large volumes of antecedent debris-flow deposits extending into the Cozy Dell reach (Figure 4) indicate that some debris flows were large enough to transfer coarse sediment downstream, but not to evacuate all of it out of the mountain catchments. The ages of these valley-filling deposits are unknown, but the boulders within them are generally unweathered, and we observed sequences of up to three debris-flow units in the widest, lowest-gradient mountain reaches of San Ysidro Creek on gradients of 0.1–0.15. The deposits imply debris flows generated by rainstorms, with or without wildfires that did not drive catchment-wide slurry production of the same magnitude as the 2018 rainstorm, which occurred only 3 weeks after the Thomas Fire (Alessio et al., 2021). Smaller debris flows or floods transported ∼1 m sized boulders onto the upper fans of Cold Spring, San Ysidro, and Romero creeks during post-wildfire rainstorms in 1964–1965 and 1969–1971 (Lancaster et al., 2021). However, the range-wide scouring of valley-filling sequences observed in 2018 appears to require flows of the same general magnitude and rheological characteristics of the 2018 event, although the minimum conditions for such a range-scale scouring event have not yet been defined.

These results suggest that the volumes of the most destructive flows arriving at the piedmont, such as the 2018 event, depend not only on the magnitude of the triggering hydrometeorological (fire and rainstorm) event but also on the accumulation of boulder-rich sediments in small fractions of the landscape. The loading of debris-flow sediment into mountain channel networks is a stochastic process, resulting from combinations of rainstorms and wildfires (Benda & Dunne, 1987, 1997). Observed and modeled probability distributions of the magnitude and frequency of these drivers in other regions (e.g., Benda et al., 1998) predict relatively frequent small debris flows that tend to deposit in steep, low-order channels, while rare large flows travel farther and deposit sediment on lower gradients (Benda & Cundy, 1990). Rheological properties of debris flows also have distributions that affect the gradients on which they come to rest, and the distances they can travel (Whipple & Dunne, 1992), and variations in valley widths and gradients can force deposition of slurry by lowering strain rates and allowing lateral drainage of pore water (Major, 1997).

The observation that antecedent accumulations of colluvial and debris-flow deposits within canyon networks have such profound effects on the volume and granulometry of debris flows may not be necessary for analyses of long-term evacuation of rock from mountain ranges where average flux rates and erosion laws may suffice (e.g., Stock & Dietrich, 2006). However, it complicates predictions of potential volumes of individual debris flows or runout modeling (Bennett et al., 2014; Iverson et al., 1998; Jakob et al., 2005; Keaton & Lowe, 1998; National Research Council, 1996). Since the volumes of sediment in storage are created by stochastic sequences of accumulation and of sufficiently large scouring events, the time since the last network-clearing flow becomes an important metric of hazard. Even neighboring catchments can differ strongly in this regard. The lidar differencing results suggest that the catchments studied now store at least 470,000 m^3 less boulder-rich sediment than before 2018. It will probably take a long time to recover those volumes fully, and replenishment could be affected by fire-rainstorm sequences, climatic fluctuations, seismicity, and (in other regions) timber harvest (Jakob et al., 2005; Swanston & Swanson, 1976). In the meantime, smaller volumes of sediment have been left behind in the form of at least 85,000 m^3 of patchy 2018 levees (Figures 1b–1d) and sparse, large boulders that became stranded by the flows, together with residual antecedent deposits which survived this event (see Figures 4d–4g).

Our observations only months after the 2018 flows suggest that boulders have already begun to collapse from the lateral residual debris-flow deposits into the center of the scoured channels. Also, some small unsorted, boulder-rich debris-flow deposits from the 2018 event already block first-order channels. They appear to have been generated from landslides that developed late in the rainstorm due to rainwater penetrating locally deep colluvium in bedrock hollows on sandstone units. We interpret that their relatively small water content and high boulder

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loads caused them to stop in low-order channels after the main debris flow had ceased. A rainstorm mobilizing the same slurry discharge from catchment hillslopes as the 2018 event could generate a debris flow that would be accommodated within the scoured channel and would encounter many fewer mobile boulders. Thus, it would not grow to the volume or peak discharge of the 2018 event. It would likely generate a shallower debris flow with a smaller boulder content, until more boulders accumulate in locations that are vulnerable to transport. The accumulation can be interrupted to various degrees by floods or debris flows of intermediate magnitude, which can scour and transport small (relative to the 2018 event), but still hazardous, volumes of boulders to the piedmont fans. In these catchments, debris flows carrying relatively small (but still damaging) volumes of boulders to the Montecito piedmont fans occurred in 1914 (without fire), 1964 (after wildfire), and 1969 (without wildfire) (Lancaster et al., 2021).

This irregular valley-filling trajectory occurring over decades to centuries suggests that process-based hazard analyses of post-wildfire debris-flow magnitude will need to include field surveys of debris accumulation upstream of high-value localities, such as settlements or infrastructure. In the absence of recent debris-flow activity, these surveys, which do not need to be precise, could be accomplished through field estimates of stored sediment volumes along sample reaches, together with the development of scaling relationships with variables such as drainage area, stream order, or gradient. Keaton (1989), Keaton and Lowe (1998), and Jakob et al. (2005) summarize methods for augmenting hydroclimatological probability analysis with geomorphological surveys to improve estimates of debris-flow probability and magnitude. Where channels have recently been scoured, and where colluvial yields are high, decadal lidar or photogrammetric surveys, updated after wildfires (e.g., DiBiase & Lamb, 2020; Ellett et al., 2019), could facilitate mapping of accumulation volumes and locations and the refinement of debris-flow magnitude predictions (De Haas et al., 2020).

Data Availability Statement

The lidar point cloud data from 2015 are archived at https://doi.org/10.25349/D9XG8S. Thomas Fire perimeter was retrieved from https://rmgsc.cr.usgs.gov/outgoing/GeoMAC/2017_fire_data/California/Thomas/.

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